

A Hash Time Lock Based on Double Encryption Algorithm

1st Ling Cao

School of Software Engineering

Chongqing University of Posts and Telecommunications

ChongQing, China

caoling@cqupt.edu.cn

2nd Zongyuan Liu

School of Software Engineering

Chongqing University of Posts and Telecommunications

ChongQing, China

ym77788@126.com

Abstract—With the rapid development of blockchain technology in recent years, many blockchain networks suitable for different types and scenarios have been born at home and abroad. Due to the characteristics of blockchain independence and network heterogeneity, there has been a scenario where data and assets cannot be exchanged and transferred between different blockchain networks, forming many value islands. Blockchain cross chain technology is an important technical means to realize the data circulation and value exchange between chains. As a mainstream cross chain technology, hash time lock (HTCL) technology originally came from Bitcoin's lightning network, which places transactions under the chain, greatly improving the speed of transactions. However, this algorithm has problems such as low security and transaction congestion in actual scenarios. This paper proposes a hash time lock based on dual encryption technology to ensure the security of the hash lock in the transmission process and effectively prevent the possibility of being intercepted and tampered with.

Keywords—cross chain; htcl; blockchain; smart contract

I. INTRODUCTION

Blockchain technology is a new technology composed of distributed databases, cryptography, point-to-point transmission mechanism, consensus algorithm, etc [1]. It truly realizes the transaction in the open p2p network world without relying on a third party. It has the characteristics of decentralization, tamper proof, traceable source data, and trustworthiness [2]. These characteristics have caused wide attention to blockchain technology, which is considered to be a change to traditional technology. It is the infrastructure of the next generation of computer technology and has attracted the attention of domestic and foreign academic circles, investment circles, and government departments. However, with the rise of various blockchains, due to the different underlying algorithm models, architectures and consensus mechanisms used by different blockchains, these blockchains are independent of each other, forming a data island [3]. In the age of big data, data contains a lot of value, so solving the data and asset circulation before different blockchains have become the focus, and cross chain technology has been born.

Cross chain technology is to enable two homogeneous or heterogeneous blockchain networks to transfer or exchange assets through specified communication protocols by writing smart contracts [4]. Cross chain types are usually asset exchange and asset transfer. Asset exchange refers to the equivalent exchange between the value of one chain and the

value of another chain [5]. Asset transfer refers to the transfer of assets in a chain to a designated chain through cross chain technology. After asset exchange, the total value of each chain will not change. Asset transfer will increase or decrease the total value of a chain. The transaction process needs to meet the atomicity and consistency of the transaction to ensure value conservation.

The current mainstream cross chain technology is divided into four types according to the algorithm and implementation mode: notary mechanism, side chain/relay, hash locking and distributed private key control. Hash locking technology is used in lightning network, Interleader and lightning network [6].

Based on the hash time lock technology, this paper proposes an improved scheme for the security problems of the algorithm. The way that the hash lock propagates directly in the channel may be hijacked and tampered by attackers. For security reasons, we design a double encryption transmission scheme, which strictly encrypts the hash lock and ensures the security of the transaction. It prevents transaction failure caused by attacks and blocks the transaction channel, and also ensures the security of user assets.

II. RELATED WORK

A. Notary Mechanism

The idea of the notary mode is similar to that of traditional exchanges. As an intermediate role, the notary is used to verify and forward messages. The system selects one or more trusted nodes as the notary, automatically or on request monitors events from different chains, and checks whether the nodes can reach consensus through specific consensus algorithms [7]. The feature of this mechanism is that users do not need to pay attention to the structural differences and consensus characteristics between chains, and this mechanism supports multiple types of cross chain asset transfers. At present, the notary mechanism can be divided into single signature notary mechanism, multi signature notary mechanism and distributed signature notary mechanism according to the different ways of signature implementation.

B. Hash Time Lock

The full name of hash lock technology is hash time lock contract. It first appeared in the Lightning Network and is the core technology for asset trading. Its principle is that without a notary, asset exchange can be completed without a third party,

and cross chain asset transactions can be realized through hash locks and time locks respectively. The basic process is as follows: Account X on chain A randomly generates a number as the key, calculates the hash value of the number as the public key, uses the public key to encrypt the assets to be exchanged, and sends the public key to account Y on chain B. Both parties encrypt their own assets through the public key and set their own time locks, and then account X transfers the key S to account Y. Account Y uses S to conduct asset transactions within the time specified for the counterparty. If the transaction succeeds, account X immediately uses the key to obtain the assets on chain B. If one party has a timeout transaction, the transaction mark fails, and the assets need to be returned to the original account.

III. SOLUTION

In the second section, we introduced the implementation principle of HTLC algorithm. The exchange of cross chain assets is realized by adding time locks and hash locks to the assets, and the asset is unlocked under appropriate conditions to obtain the opposite party's assets. However, the reliable delivery of locks is a prerequisite for successful transactions. But through research, if the locks are damaged or tampered with by attackers, they will cause transaction congestion or asset loss. When the transaction starts, the transaction initiator will generate a random number S, and then use the hash function to calculate the hash value of S. After the asset is locked with the hash value, the hash value needs to be transmitted to the payee, who also uses the hash value to lock the asset. Then the hash value may be intercepted and tampered during the transmission process. If the attacker modifies the hash value, the payee will lock the asset with the wrong hash value, which will lead to frequent transaction failures or network congestion.

Aiming at this security problem, this paper proposes an improved scheme, which uses double salt value encryption transmission and digital signature technology to ensure the security of hash lock transmission. The scheme is as follows:

- 1) First, user Alice initiates an asset exchange transaction from Chain A to Bob on Chain B. Alice generates a random number S
- 2) Calculate the hash value of S $h = \text{hash}(S)$, use h to lock Alice's assets, specify a time point T1, and use T1 to add a time lock to the assets
- 3) Then randomly generate a salt value, merge the salt and h according to some rules to get the result X, and then use the MD5 encryption algorithm to double encrypt X1 to get the result Hx
- 4) Both Hx and h are transmitted to Bob, and the salt value is encrypted using the public key provided by Bob and transmitted to Bob

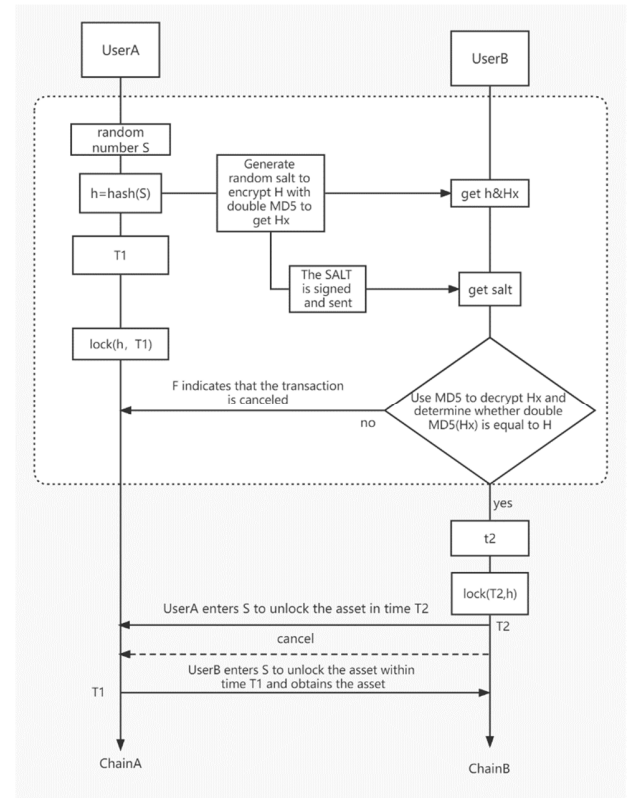


Figure 1. Schematic diagram of improved algorithm

- 5) When Bob receives the encrypted salt, he uses the private key to decrypt it to get the original value salt, and then uses the salt and MD5 algorithms to verify the received hash value h.
- 6) If the calculated result R is equal to Hx, it means that the received value of h has not been tampered with. Bob uses h to lock the asset, and specifies time point T2 to add time lock to the asset
- 7) If R is not equal to Hx, it indicates that the original value h has changed during the network transmission process and the transaction is unsafe, then notify the other party to terminate the transaction.

The improved algorithm principle is shown in Figure 1. Table 1 and Table 2 are pseudocodes of the initiator's transaction algorithm and the receiver's transaction algorithm respectively.

Table 1 describes the transaction algorithm of the initiator. The input parameters of this method mainly include target account, transaction asset and lock time. First, judge whether the current asset of the user is less than the transaction asset. If it is less than the transaction asset, the transaction will be ended directly. Otherwise, the algorithm will continue to execute. Use the getrandom method to obtain a random value S, then use the hash algorithm to generate the corresponding hash value, call the lock method, and pass in the h and T1 parameters to lock the asset. At this point, locking is complete. Next, call the generateSaltByTimestamp method, generate the salt according

to the current timestamp, and use the public key to encrypt the salt to generate an esp. Then, transfer h, Hx, and esp to the target to get the result res. Judge whether the transfer is successful according to res. If it is successful, continue to execute the transaction, call the TryAcquireTargetAsset method to exchange assets, and tell the user whether the transaction is successful according to the return value.

Table 2 describes the receiver's transaction algorithm. This method mainly has five parameter values, namely, h, Hx, esp, asset, and T1. In this method, the first is to call the decodeByPrivateKey method, which uses the private key to decrypt the esp value to get salt, and then transforms and encrypts the salt twice to get res, to determine whether res is equal to the Hx passed. If not, It means that the data has been tampered or stolen in the transmission process, and the transaction failure information is returned.

TABLE I: ORIGINATOR TRADING ALGORITHM

Algorithm 1 Originator trading algorithm
input: targetAccount, aaset, T1
output: isSuccess
<pre> function ExchangeAssetWithTargetAccount (targetAccount,asset,T1) if(currentAsset < asset) return end S ← GetRandom() h ← hash(S) lockCurrentAssetByTimelockAndHashValue(h,T1) salt ← generateSaltByTimestamp() esp ← encryptSaltWithPublicKey(salt) res ← transferToTarget(h,Hx,esp)//transfer to target if(!res) return fail isSuccess ← TryAcquireTargetAsset(targetAccount,h) if (isSuccess) return success else return fail </pre>

TABLE II: RECEIVER TRADING ALGORITHM

Algorithm 2 Receiver trading algorithm
input: h, Hx, esp, account, aaset, T1
output: isSuccess
<pre> function acceptAssetWithTargetAccount(h,Hx,esp, asset,T1) salt ← decodeByPrivateKey(esp) res ← MD5(salt+ MD5(salt+h)) if(res!=Hx) return fail T2 ← generateTime LockCurrentAssetByTimeLockAndHashValue(T2,h) isSuccess ← TryAcquireTargetAsset(account,h,T1) if (isSuccess) return success else return fail </pre>

Otherwise, the transaction continues to be executed. Call the generateTime method to generate a time lock T2, and use the lock method to pass in T2 and h to lock the current asset. Then, execute the TryAcquireTargetAsset method to try to conduct the transaction, and get the return value. Finally, judge whether the transaction is successful according to the return value.

IV. EXPERIMENTAL ANALYSIS

In order to evaluate the performance of the improved algorithm, we will conduct experimental analysis of the algorithm in this section.

A. Experimental environment

This experiment uses the Ethereum platform to build the experimental environment. As one of the mainstream blockchain projects, Ethereum has a stable and good experimental environment. The experiment uses two servers to build Ethereum nodes, each configured with a 4-core Intel Core Processor (Skylake, IBRS) CPU, 8GB memory, 500GB storage space, and 1000M bandwidth network.

B. Algorithm Performance Analysis

The security of cross chain asset exchange is crucial, and it is the core indicator of the evaluation algorithm. The security of cross chain is bound to be closely related to the problems faced by the blockchain in the actual scenario. This paper mainly conducts security experimental analysis on the improved HTLC algorithm, mainly analyzing the hash collision of the algorithm. In the process of cross chain asset exchange, it is necessary to calculate the required hash value through the hash function, and use the hash value to lock the asset. The security of the transaction depends on the security of the hash value here. The attacker uses violent strong collision to calculate the same hash value to achieve the purpose of camouflage, and successfully connects with the target user and obtains the other party's assets. In this section, experiments are carried out by collision attack and original image attack, and mainstream hash algorithm is used. The experimental results are shown in Figure 2.

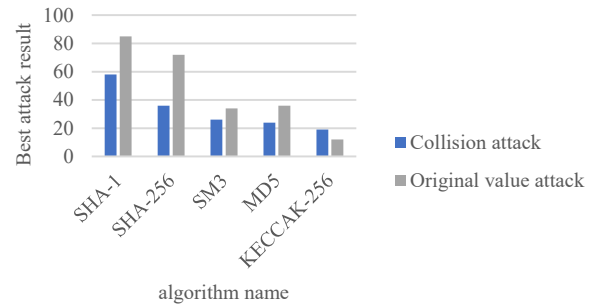


Figure 2. Security comparison of algorithms

The experimental results show that the collision attack and preimage attack of KECCAK-256 algorithm are 18.6% and 12.2%, respectively, and the performance effect is the best among the five algorithms. The collision attack and preimage attack of SHA-256 algorithm are 36.6% and 71.8%, respectively. The collision attack and preimage attack of SHA-1 algorithm are 58.5% and 84.6% respectively, which are the worst among the five algorithms and bring great risks to asset trading. At present, attackers have formed effective collision attacks on SHA-1 algorithms, but there is no scheme for collision attacks on strong algorithms such as SM3 in theory. Attackers can only use brute force cracking to attack, but the attack cost will be high.

C. Algorithm Performance Analysis

The performance of HTLC algorithm is mainly determined by the execution time of the algorithm, and the calculation formula is as follows:

$$T_{\text{execute}} = T_{\text{begin}} + T_{\text{consensus}} + T_{\text{end}} \quad (1)$$

T_{begin} indicates the start time of asset conversion, $T_{\text{consensus}}$ represents the consensus time of the transaction on the chain, T_{end} indicates the end time of asset conversion. In this experiment, we set the upper limit of the number of transactions as 5000, and compare the difference between the original HTLC algorithm and the improved back-end HTLC algorithm in terms of transaction performance in a ladder like manner, The experimental results are shown in Figure 3.

According to the experimental results, the transaction time of the original HTLC algorithm shows an obvious upward trend after 1000 transactions. When the transaction volume reaches about 5000, the time consumption has reached 500ms. The performance of the improved HTLC algorithm is obviously better than the original algorithm, and the execution time consumption shows a slowly increasing trend. After the transaction volume is more than 2000, the transaction time is greatly reduced, which improves the transaction volume of the system and enables users to complete more cross chain asset

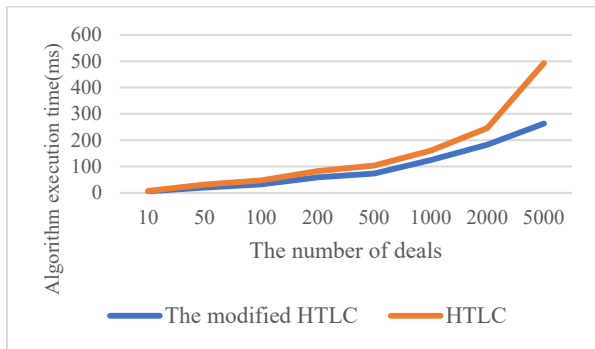


Figure 3. Algorithm performance analysis

transactions in a certain time. Because this scheme guarantees the execution efficiency of the transaction, optimizes the algorithm, and reduces the transaction failure and congestion caused by external attacks, the experimental results prove the effectiveness of this improved scheme.

V. CONCLUSIONS

In this paper, the current mainstream cross-chain technology is analyzed and summarized. By comparing various cross-chain technology principles, the HTLC algorithm is deeply studied, and an improved scheme of double hash lock encryption in the transmission process is proposed. The improved algorithm is defined and implemented. Finally, the experiment is carried out and the experimental results are analyzed. By comparing with the original HTLC algorithm, the cross-chain algorithm proposed in this paper has certain advantages in security and performance, which further expands the application of the algorithm in the actual scene.

ACKNOWLEDGMENT

This work was funded by the Chongqing Municipal Education Commission Science and Technology Research Project under grant agreement number KJQN201900646.

REFERENCES

- [1] S. Wang, L. Ouyang, Y. Yuan, X. Ni, X. Han and F. -Y. Wang, "Blockchain-Enabled Smart Contracts: Architecture, Applications, and Future Trends," in *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 49, no. 11, pp. 2266-2277, Nov. 2019.
- [2] L. Bahri and S. Girdzijauskas, "Blockchain Technology: Practical P2P Computing (Tutorial)," *2019 IEEE 4th International Workshops on Foundations and Applications of Self* Systems (FAS*W)*, Umea, Sweden, 2019, pp. 249-250.
- [3] M. N. M. Bhutta *et al.*, "A Survey on Blockchain Technology: Evolution, Architecture and Security," in *IEEE Access*, vol. 9, pp. 61048-61073, 2021.
- [4] S. Lin, Y. Kong and S. Nie, "Overview of Block Chain Cross Chain Technology," *2021 13th International Conference on Measuring Technology and Mechatronics Automation (ICMTMA)*, Beihai, China, 2021, pp. 357-360.
- [5] Monika, A. Goyal, S. Raina, K. Bhatia and R. Bhatia, "Atomic Cross-Chain Asset Exchange for Ethereum Public Chains," *2021 International Conference on Computer Communication and Informatics (ICCCI)*, Coimbatore, India, 2021, pp. 1-4.
- [6] J. Rueegger and G. S. Machado, "Rational Exchange: Incentives in Atomic Cross Chain Swaps," *2020 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, Toronto, ON, Canada, 2020, pp. 1-3.
- [7] K. Honari *et al.*, "A Scalable Blockchain-based Smart Contract Model for Decentralized Voltage Stability Using Sharding Technique," *2022 IEEE International Conference on Blockchain (Blockchain)*, Espoo, Finland, 2022, pp. 124-131.